# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## ANSWER KEY & MARKING SCHEME SET A

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Mathematics | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 12 × 1 = 12

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 250 cm² | $2(50+25+50)$ |
| 2 | **(c)** the base of the hemisphere and the top of the cylinder | Faces in contact are hidden |
| 3 | **(b)** Student A is correct | No flat base is exposed |
| 4 | **(c)** 220 cm² | $\frac{22}{7}\times 7\times 10$ |
| 5 | **(c)** 26 | $3(22)-2(20)$ |
| 6 | **(c)** the sum of frequencies up to and including that class | Definition |
| 7 | **(b)** Incorrect — use $l+\frac{\frac{n}{2}-cf}{f}\times h$ | Median needs the formula |
| 8 | **(c)** 2 | $\frac{45-25}{10}$ |
| 9 | **(c)** 0.99 | $0\le P\le 1$ |
| 10 | **(d)** 0.65 | $1-0.35$ |
| 11 | **(b)** $\frac{1}{2}$ | HT, TH out of 4 |
| 12 | **(b)** Incorrect; $P(\text{not } A)=0.3$ | $P(A)+P(\bar{A})=1$ |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 13 | $1540+\frac{2}{3}\times\frac{22}{7}\times 343=1540+718.67=2258.67$ cm³ | 1+1 |
| 14 | Volume $=54$ cm³ (claim false). Cuboid $6\times 3\times 3$: SA $=2(18+9+18)=90$ cm² | 1+1 |
| 15 | Modal 40–60: $40+\frac{18-6}{36-6-10}\times 20=52$ | 1+1 |
| 16 | $\Sigma fx=30+100+210+360=700$, $\Sigma f=20\Rightarrow$ mean $=35$. Claim is false (frequencies ignored). | 1+1 |
| 17 | (i) HT, TH, TT ⇒ $\frac{3}{4}$ (ii) $\frac{1}{4}$ | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 18 | Diameter 7 cm. SA $=6\times 49-\pi r^2+2\pi r^2=294+38.5=332.5$ cm² | 1+2 |
| 19 | $l=2.5$: cylinder CSA $=10.56$, cone CSA $=5.5$, base $=1.54\Rightarrow 17.6\approx 18$ cm² | 1+1+1 |
| 20 | Modal class 35–45: $35+\frac{23-21}{46-21-14}\times 10=35+\frac{20}{11}=36.82$ years | 1+1+1 |
| 21 | (i) Green $=16$, blue $=8$ (ii) $\frac{16}{28}=\frac{4}{7}$ | 2+1 |
| 22 | $2\pi r^2+2\pi rh=2\times\frac{22}{7}\times 10.5(10.5+14)=1617$ cm² | 1+1+1 |
| | **OR:** $198-\frac{264}{7}=\frac{1122}{7}\approx 160.29$ cm³ | 1+1+1 |
| 23 | $n=50$; cf 6, 16, 30 ⇒ class 20–30: $20+\frac{25-16}{14}\times 10=26.43$ | 1+1+1 |
| | **OR:** $a=225,\ h=50$; $\Sigma fu=-8-5+0+2+4=-7$; mean $=225-\frac{350}{25}=$ ₹211 | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 24 | (i) 17.5 m (ii) $264+770=1034$ m² (iii) ₹82,720 (iv) $1848+2156=4004$ m³ | 1+2+1+1 |
| | **OR:** (i) 141.3 cm³ (ii) 56.52 cm³ (iii) 197.82 cm³ (iv) 9891 cm³ $\approx 9.89$ L | 1+1+2+1 |
| 25 | (i) $\frac{6}{30}=\frac{1}{5}$ (ii) 1,4,9,16,25 ⇒ $\frac{1}{6}$ (iii) $6+5-1=10\Rightarrow\frac{1}{3}$ (iv) 25 is counted twice | 1+1+2+1 |
| | **OR:** (i) $\frac{18-x}{18}$ (ii) $\frac{x+2}{20}=\frac{9}{8}\cdot\frac{x}{18}=\frac{x}{16}\Rightarrow 16x+32=20x\Rightarrow x=8$ (so P(not red) $=\frac{5}{9}$) | 2+3 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET B

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Mathematics | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 12 × 1 = 12

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** 179.67 cm³ | $r = 3.5$: $\frac{4}{3} \times \frac{22}{7} \times 42.875$ |
| 2 | **(b)** 1540 cm³ | $\frac{22}{7} \times 49 \times 10$ |
| 3 | **(d)** $2\pi r^2$ | Two new circular faces |
| 4 | **(b)** $3\pi r^2$ | Curved $2\pi r^2$ + base $\pi r^2$ |
| 5 | **(d)** 54 | $50 + 10 \times 0.4$ |
| 6 | **(d)** 20–30 | Highest frequency 12 |
| 7 | **(a)** mode | Most frequent value |
| 8 | **(a)** Student A is correct | B described the modal class |
| 9 | **(c)** $\frac{1}{26}$ | $\frac{2}{52}$ |
| 10 | **(c)** 1 | All outcomes favourable |
| 11 | **(c)** 115% | $115\% > 1$ |
| 12 | **(b)** Incorrect — $0 \le P(E) \le 1$ | Probability is never negative |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 13 | $l = 10$: $\pi rl = 188.4$; $2\pi r^2 = 226.08$ ⇒ 414.48 cm² | 1+1 |
| 14 | $r = 3.5$, cylinder 14 mm: $2\pi rh = 308$, $4\pi r^2 = 154$ ⇒ 462 mm² | 1+1 |
| 15 | $n = 60$; cf 6, 15, 35 ⇒ median class 40–60; modal class 40–60 | 1+1 |
| 16 | Total 180. (i) $\frac{100}{180} = \frac{5}{9}$ (ii) $\frac{170}{180} = \frac{17}{18}$ | 1+1 |
| 17 | False. The two outcomes are not equally likely, so the classical formula cannot be used. | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 18 | $\frac{1}{3} \times \frac{22}{7} \times 12.25 \times 12 + \frac{2}{3} \times \frac{22}{7} \times 42.875 = 154 + 89.83 = 243.83$ cm³ | 1+1+1 |
| 19 | $a = 150$; $d = -40, -20, 0, 20, 40$; $\Sigma fd = -480 - 280 + 0 + 120 + 400 = -240$; mean $= 150 - \frac{240}{50} =$ ₹145.20 | 1+1+1 |
| 20 | 25 outcomes. (i) $\frac{5}{25} = \frac{1}{5}$ (ii) 8 outcomes ⇒ $\frac{8}{25}$ (iii) $1 - \frac{1}{5} = \frac{4}{5}$ | 1+1+1 |
| 21 | $\frac{x}{5+x} = 2 \times \frac{5}{5+x} \Rightarrow x = 10$ blue balls | 1+1+1 |
| 22 | Modal 30–40: $30 + \frac{16-x}{32-x-12} \times 10 = 36 \Rightarrow \frac{16-x}{20-x} = 0.6 \Rightarrow x = 10$ | 1+1+1 |
|  | **OR:** $p + q = 20$; $2060 + 30p + 70q = 3140 \Rightarrow 3p + 7q = 108$ ⇒ $q = 12,\ p = 8$ | 1+1+1 |
| 23 | (i) $\frac{10}{35} = \frac{2}{7}$ (ii) $\frac{5}{7}$ (iii) $\frac{15}{40} = \frac{3}{8}$ | 1+1+1 |
|  | **OR:** (i) $\frac{50}{100} = \frac{1}{2}$ (ii) 4, 9, …, 100: 9 ⇒ $\frac{9}{100}$ (iii) 2,3,5,7,11,13,17,19 ⇒ $\frac{8}{100} = \frac{2}{25}$ | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 24 | (i) $2\pi r(h + r) = 2 \times \frac{22}{7} \times 30 \times 175 = 33000$ cm² $= 3.3$ m² (ii) $\frac{2}{3} \times \frac{22}{7} \times 27000 \approx 56571$ cm³ $\approx 56.57$ L | 3+2 |
|  | **OR:** (i) $1600 + \frac{1}{2} \times \frac{22}{7} \times 25 \times 20 \approx 2385.71$ m³ (ii) $\pi rh = \frac{22}{7} \times 5 \times 20 \approx 314.29$ m² | 3+2 |
| 25 | Frequencies 3, 5, 9, 3, 3. Median: $\frac{n}{2} = 11.5$ ⇒ $20 + \frac{3.5}{9} \times 10 \approx 23.89$ (3). Mean $= \frac{555}{23} \approx 24.13$ (2) | 3+2 |
|  | **OR:** (i) Modal 0–10: $0 + \frac{22}{44-10} \times 10 \approx 6.47$ mm (ii) $\frac{1800}{70} \approx 25.71$ mm (iii) Wrong — the highest frequency is in 0–10 | 2+2+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET C

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Mathematics | Maximum Marks: 50

### SECTION — A (Objective Type) — Answers — 12 × 1 = 12

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 66 cm³ | $\frac{1}{3} \times \frac{22}{7} \times 9 \times 7$ |
| 2 | **(a)** Student A is correct | $440 + 154 + 308 = 902$; B counted the hidden top face |
| 3 | **(b)** Incorrect — the surfaces in contact are not counted | Joined faces become hidden |
| 4 | **(d)** 3 : 1 | $\pi r^2 h : \frac{1}{3}\pi r^2 h$ |
| 5 | **(a)** 48 | $\frac{24+2(60)}{3}$ |
| 6 | **(d)** 20–30 | $25 \pm 5$ |
| 7 | **(a)** $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$ | Standard formula |
| 8 | **(b)** 5.5 | $\frac{55}{10}$ |
| 9 | **(c)** 0 | No favourable outcome |
| 10 | **(a)** 1 | Sure event |
| 11 | **(d)** $\frac{6}{13}$ | A, A, A, I, I, O: 6 of 13 |
| 12 | **(b)** $\frac{1}{13}$ | $\frac{4}{52}$ |

### SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 13 | $\frac{880}{7} + \frac{88}{7} = \frac{968}{7} \approx 138.29$ cm³ | 1+1 |
| 14 | Modal class 40–50: $40 + \frac{28-12}{56-12-20} \times 10 = 40 + \frac{16}{24} \times 10 = 46.67$ | 1+1 |
| 15 | $\frac{285+15p}{18+p} = 15.5 \Rightarrow 0.5p = 6 \Rightarrow p = 12$ | 1+1 |
| 16 | False. $\frac{1}{2}$ is a theoretical probability; it does not guarantee 5 heads in 10 tosses (we may get 4, 6, …). It only means heads occur about half the time in a large number of tosses. | 1+1 |
| 17 | (i) 2,3,5,7,11,13,17,19 ⇒ $\frac{8}{20} = \frac{2}{5}$ (ii) 5,10,15,20 ⇒ $\frac{4}{20} = \frac{1}{5}$ | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 18 | $264 + 55.44 + 110.88 = 430.32$ cm² | 1+1+1 |
| 19 | Frequencies 5, 7, 13, 13, 7, 5; $\frac{n}{2} = 25$ ⇒ class 20–30: $20 + \frac{25-12}{13} \times 10 = 30$ | 1+1+1 |
| 20 | $15 + \frac{20-11}{40-11-14} \times 5 = 18$ | 1+1+1 |
| 21 | (i) $\frac{4}{20} = \frac{1}{5}$ (ii) $\frac{4}{19}$ | 1+2 |
| 22 | (i) $\frac{2}{6} = \frac{1}{3}$ (ii) $\frac{1}{6}$ (iii) A, E, A ⇒ $\frac{3}{6} = \frac{1}{2}$ | 1+1+1 |
| | **OR:** White $= 6$, black $= 15$ ⇒ red $= 30 - 21 = 9$ | 1+1+1 |
| 23 | Apparent $= 3.14 \times 6.25 \times 10 = 196.25$ cm³; hemisphere $= \frac{2}{3} \times 3.14 \times 15.625 = 32.71$; actual $= 163.54$ cm³ | 1+1+1 |
| | **OR:** $6l^2 - \frac{\pi l^2}{4} + \frac{\pi l^2}{2} = \frac{l^2}{4}(24 + \pi)$ | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 24 | Median: cf 2, 10, 22, 46 ⇒ $70 + \frac{50-46}{38} \times 5 = 70.53$. Mode: $70 + \frac{38-24}{76-24-16} \times 5 = 71.94$ | 3+2 |
| | **OR:** Mean $= 25$ (2); median: cf 5, 15 ⇒ $20 + \frac{10}{20} \times 10 = 25$ (2); mode $= 20 + \frac{10}{40-20} \times 10 = 25$; all equal — symmetric data (1) | 2+2+1 |
| 25 | 46 cards left. (i) $\frac{6}{46} = \frac{3}{23}$ (ii) $\frac{26}{46} = \frac{13}{23}$ (iii) $\frac{20}{46} = \frac{10}{23}$ (iv) $\frac{2}{46} = \frac{1}{23}$ | 1+1+1+2 |
| | **OR:** (i) $\frac{x}{12}$ (ii) $\frac{x+6}{18} = \frac{2x}{12} \Rightarrow x = 3$ (iii) $\frac{9}{18} = \frac{1}{2}$ | 1+3+1 |

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